Birmingham River Champion's
Urban Outfall Safari Overview

This document adapts information shared in the the Zoological Society of London's Urban Outfall Safari guidance pack, which you should check out if you would like to find out more!

Water pollutants move from land to river through several different pathways, a key one in urban environments being drainage systems. There are two main types of drainage network, a single drainage networ or a two drainage networks.

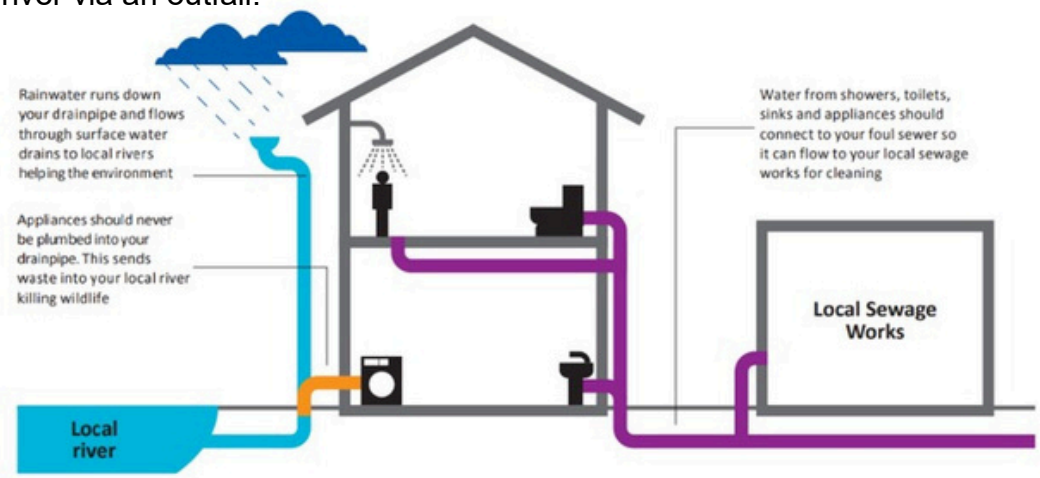
Single Pipe Networks

In regions (typically older residential areas) that are only served by a single drainage / single pipe sewage network, foul and rainfall runoff water share the same pipes and both are transferred to local sewage treatment works. However, these systems can become overwhelmed during heavy rainfall, which is when 'combined sewage overflows' discharge a mix of the two water types into river environments.

This may lead to impacts including contributing to elevated phosphate and ammonia levels. Moreover, when surface water drains are polluting, evidence such as grey ('sewage') fungus, sanitary products and discoloured water can be seen at outfalls (visible discharge points that are generally over 20cm in diameter|).

Two Pipe Networks

For regions served by two drainage / two pipe sewage networks, one set of pipes conveys foul waste (e.g., water from your toilet, shower and washing machine) to a sewage treatment works, and the other pipe sends surface water (rainfall runoff) to the nearest river via an outfall.



A schematic a house connected to two drainage networks (sourced from the Zoological Society of London's Urban Outfall Safari guidance pack).

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These outfalls are visible discharge points that are generally over 20cm in diameter. 'Misconnection' occurs when household appliances (e.g., washing machines, dishwashers, toilets) are incorrectly plumbed into the surface water drains and their wastewater ends up travelling directly into rivers via outfalls, rather than to sewage treatment facilities. Two pipe sewer networks are more common in modern housing developments.

Surface water outfalls can allow pollution into our rivers in other ways too. Blockages in the foul sewers, caused for example by a build-up of fat or wet wipes, can also pollute rivers by causing foul wastewater to back up in the sewers and drain into the surface water network and then to rivers.

What Pollution Looks Like: At A Glance

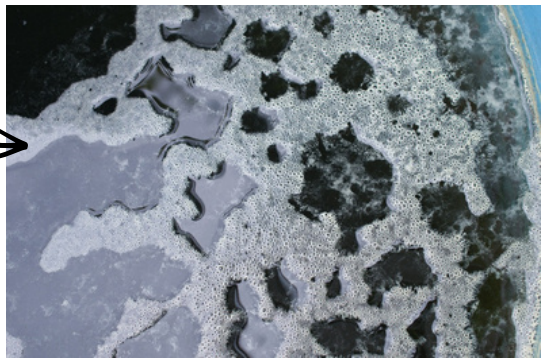
Sewage 'rag' debris (e.g., sanitary products)





Grey fungus (tufty growth on riverbed)

Surface scum





Plumes of discoloured water

'Urban Outfall Safari' allows volunteers to characterise how outfalls are polluting river environments. The technique provides a rapid assessment using a easily interpretable survey form that can be performed from the riverbank (i.e., no water access required). The survey should be undertaken during lower flow conditions as floods can wash away evidence of pollution and mask background misconnection issues. Therefore 48 hours of no or only light rain in the catchment is needed before conducting any survey work.

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Urban Outfall Safari Surveying

Urban Outfall Safari requires collecting the following information:

- The volunteers name (so that we can follow up on any information if needed)
- Location of the outfall (the name of the river and either GPS or WhatThreeWords location)
- A description of the outfall (including nearby landmarks and which bank the outfall is on)
- A photo of the outfall
- Flow coming out of the outfall
- The visual impact of the outfall
- The aesthetics of the outfall
- Description of other forms of pollution

Measuring Flow

Flow coming out of the outfall is scored from a-e, with each each score relating to the following flow:

a. No flow
b. Trickle: <0.1l/s, enough to fill a teacup in 1 minute
c. Low flow: 0.1-1l/s, enough to fill a bucket in 1 minute
d. Moderate flow: 1-2l/s, fills more than 1 bucket in 1 minute
e. High flow: >2l/s, clearly fills more than a bath tub in 1 minute

Measuring Visual Impact

Visual impact, or the distance from the outfall where pollution can be seen, is scored from a-e as follows:

a. No visible effect
b. Within 2m of outfall
c. Impact 2m to 10m from outfall
d. Impact 10m to 30m from outfall
e. Impact greater than 30m from outfall

There might not be a perfect fit, but select the category that most closely matches what is there.

Measuring Aesthetics

The aesthetics, or smell/appearance/type of pollution, is scored from a-e as follows:

a. No odour or visible aesthetics
b. Faint smell, slight discolouration
c. Mild smell, mild discolouration, small coverage of grey fungus
d. Strong smell, strong discolouration, large coverage of grey fungus and/or litter
e. Gross smell, gross sewage

Again, there might not be a perfect fit, but select the category that most closely matches what is there. See the previous page for images of what different pollution types might look like.